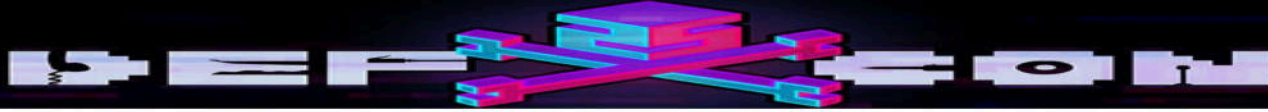


# Exploiting Old Mag-stripe Information with New Technology

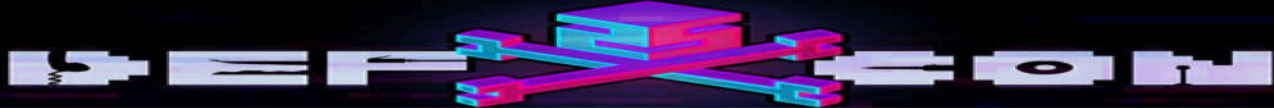
Salvador Mendoza  
Twitter: @Netxing  
Blog: [salmg.net](http://salmg.net)



## About Me

- Security Researcher
- Samsung Pay:  
Tokenized Numbers,  
Flaws and Issues





## Analyzing previous talks/tools

- **Major Malfunction DEFCON 14**

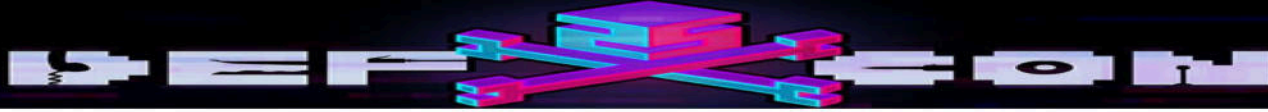
Magstripe Madness

- **Samy Kamkar**

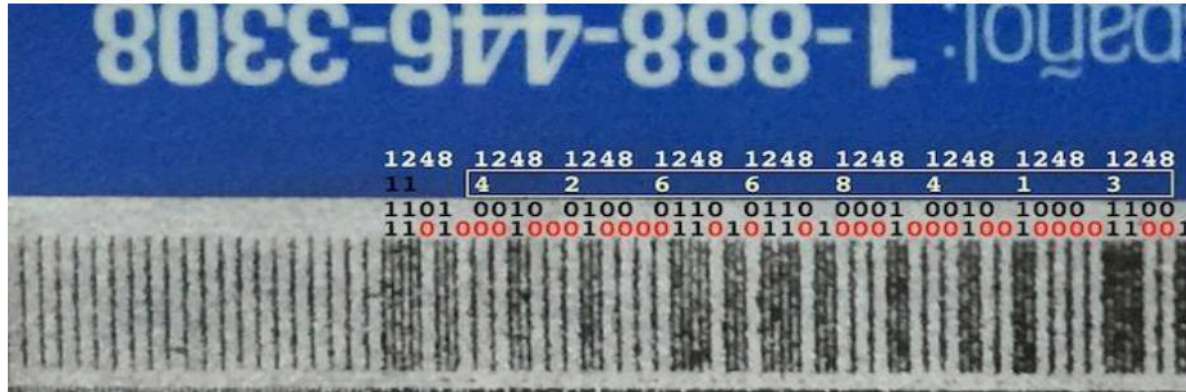
MagSpoof - 2015

- **Weston Hecker DEFCON 24**

Hacking Hotel Keys and Point of Sale Systems

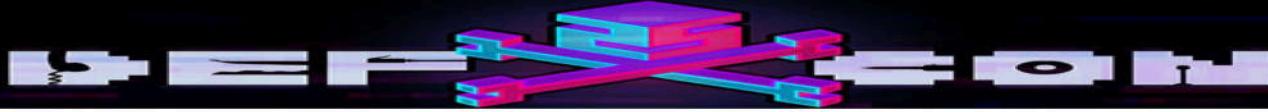


# Intro to Magnetic Stripe Information



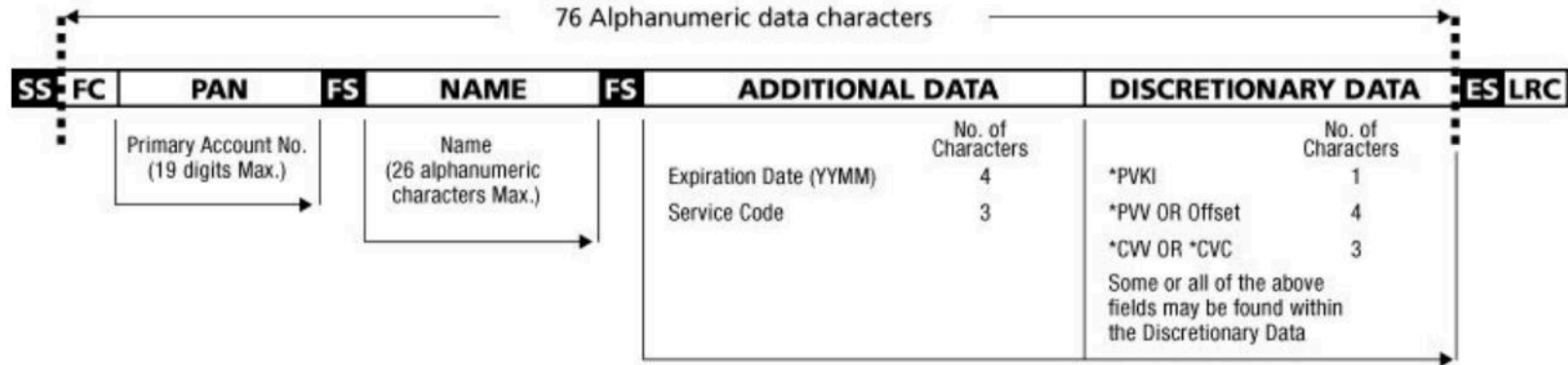
- Type of card capable of storing data by modifying the magnetism of tiny iron-based magnetic particles on a band of magnetic material on the card
- TL;DR → Track1 = [UPPERCASE,numbers] Track 2/3 = Numbers





# Magstripe Composition

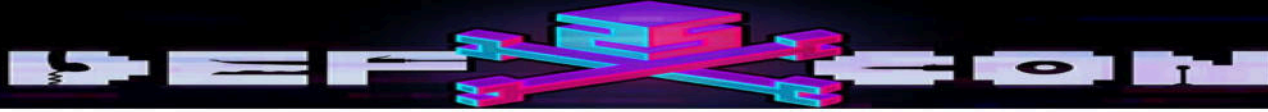
%B4929555123456789^MALFUNCTION/MAJOR ^09022010100000000000000970000000?



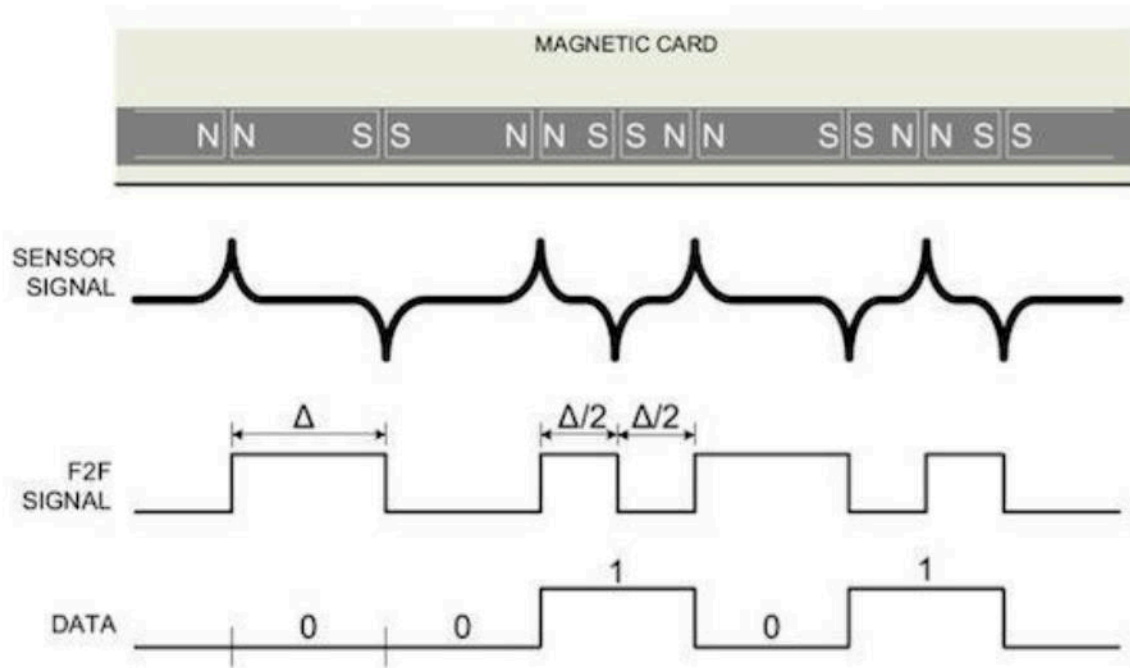
Shaded area identifies control characters

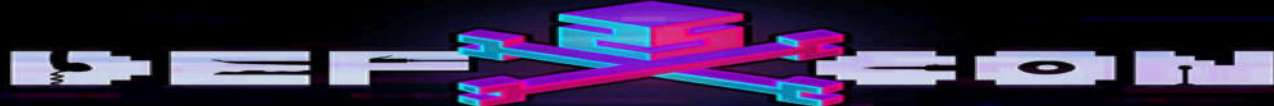
|           |                 |   |            |                                         |
|-----------|-----------------|---|------------|-----------------------------------------|
| <b>SS</b> | Start Sentinel  | % | <b>FC</b>  | Format Code                             |
| <b>FS</b> | Field Separator | ^ | <b>LRC</b> | Longitudinal Redundancy Check Character |
| <b>ES</b> | End Sentinel    | ? |            |                                         |

\* (PVKI) PIN Verification Key Indicator  
\* (PVV) PIN Verification Value  
\* (CVV) Card Verification Value  
\* (CVC) Card Validation Code



# Magstripe info, Parity, and Waves

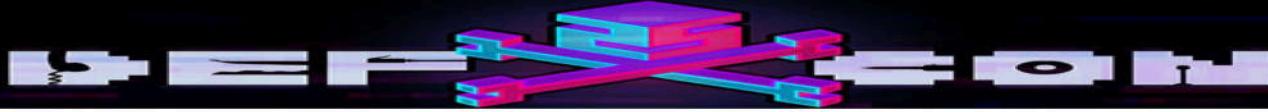




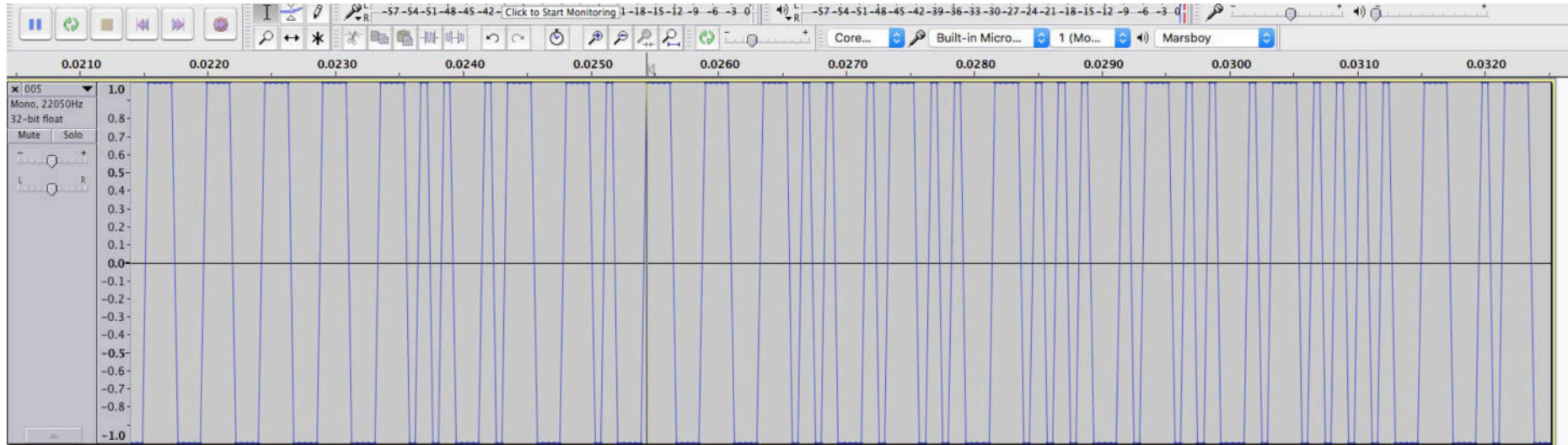
# Magstripe info, Parity, and Waves

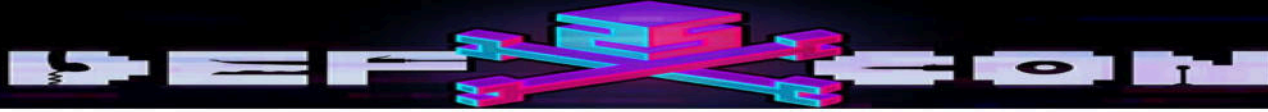
| --Data Bits-- |    |    |    | Parity | Character | Function        |
|---------------|----|----|----|--------|-----------|-----------------|
| b1            | b2 | b3 | b4 | b5     |           |                 |
| 0             | 0  | 0  | 0  | 1      | 0 (0H)    | Data            |
| 1             | 0  | 0  | 0  | 0      | 1 (1H)    | "               |
| 0             | 1  | 0  | 0  | 0      | 2 (2H)    | "               |
| 1             | 1  | 0  | 0  | 1      | 3 (3H)    | "               |
| 0             | 0  | 1  | 0  | 0      | 4 (4H)    | "               |
| 1             | 0  | 1  | 0  | 1      | 5 (5H)    | "               |
| 0             | 1  | 1  | 0  | 1      | 6 (6H)    | "               |
| 1             | 1  | 1  | 0  | 0      | 7 (7H)    | "               |
| 0             | 0  | 0  | 1  | 0      | 8 (8H)    | "               |
| 1             | 0  | 0  | 1  | 1      | 9 (9H)    | "               |
| 0             | 1  | 0  | 1  | 1      | : (AH)    | Control         |
| 1             | 1  | 0  | 1  | 0      | ; (BH)    | Start Sentinel  |
| 0             | 0  | 1  | 1  | 1      | < (CH)    | Control         |
| 1             | 0  | 1  | 1  | 0      | = (DH)    | Field Separator |
| 0             | 1  | 1  | 1  | 0      | > (EH)    | Control         |
| 1             | 1  | 1  | 1  | 1      | ? (FH)    | End Sentinel    |



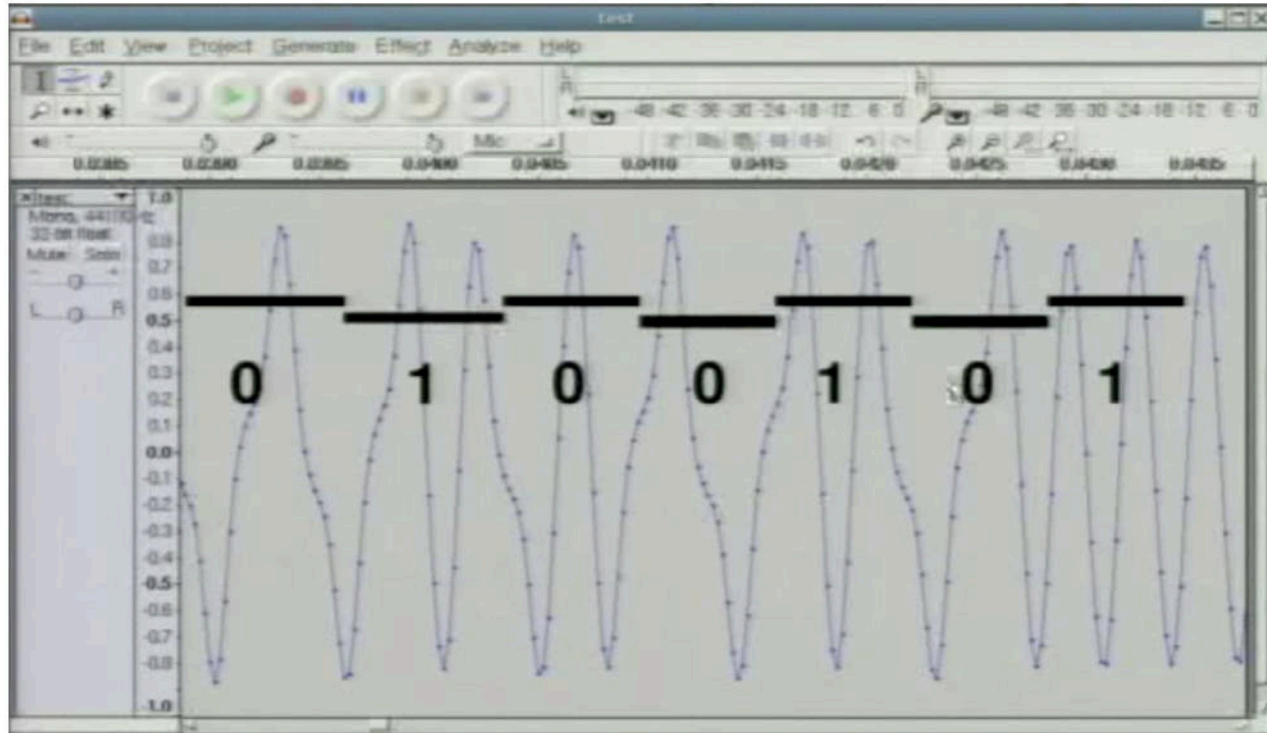


# Magstripe Signal

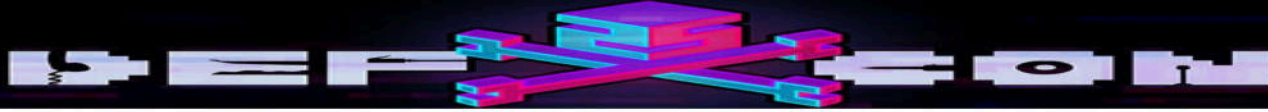




# Major Malfunction DEFCON 14



<https://www.youtube.com/watch?v=ITihB1c3dHw>



# BlueSpooof Descendancy

## MagSpooof

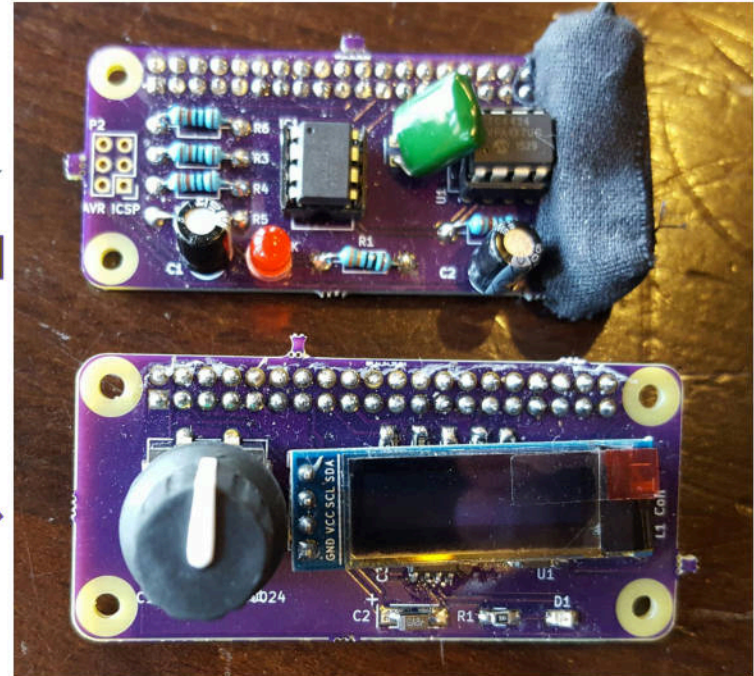


<https://www.samy.pl>

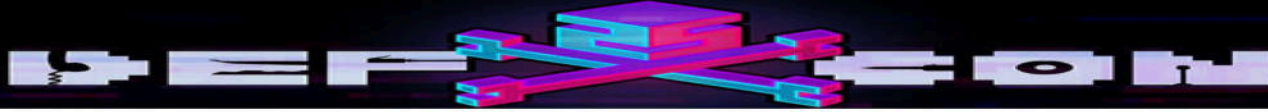
MagSpooofPI

SamyKam

## First Prototypes



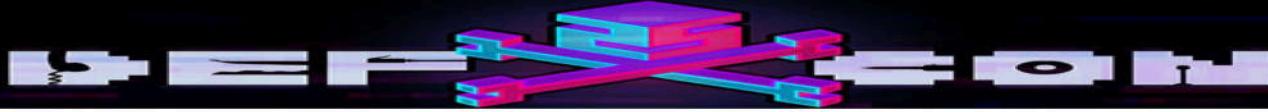
Designed PCB by @electronicats



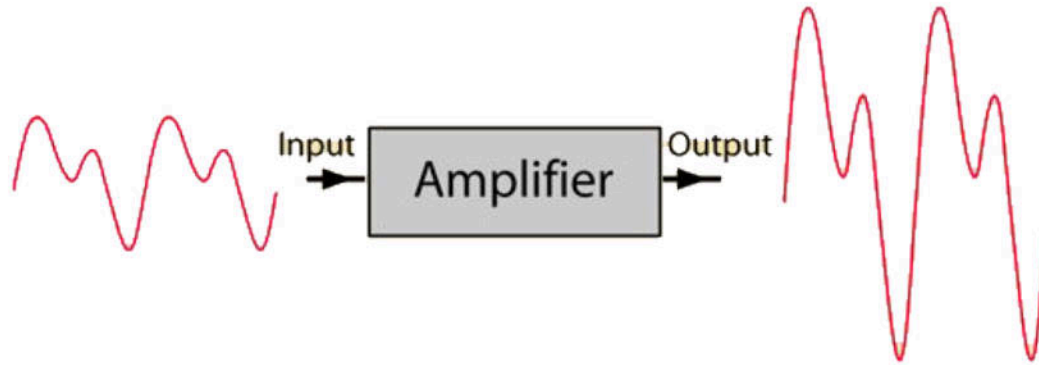
# Weston Hecker - DEFCON 24

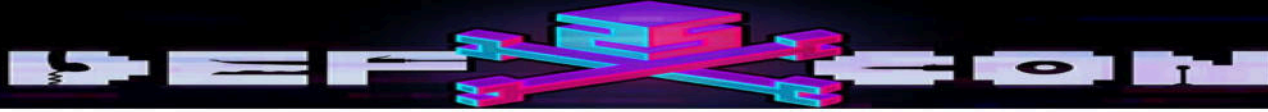




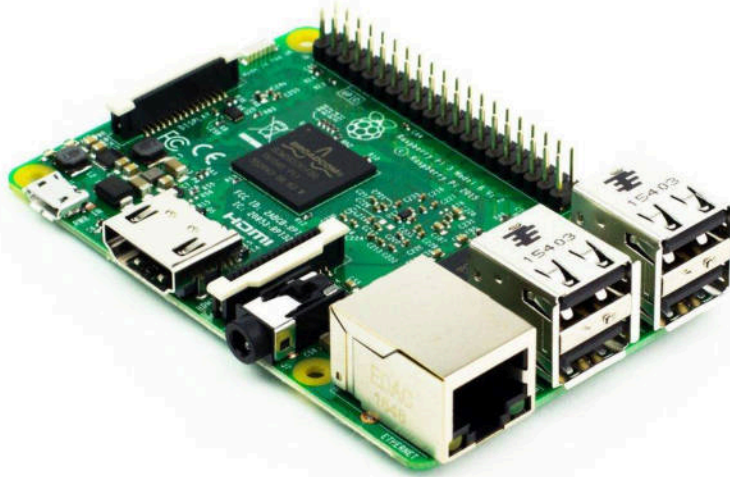


# Sound Amplifier

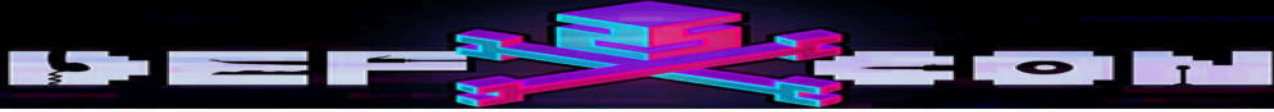




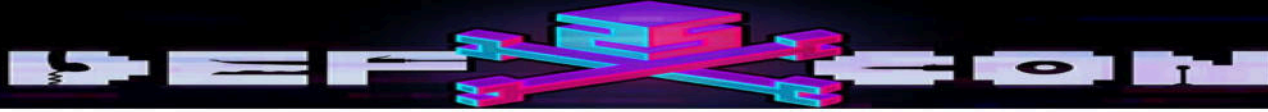
# Raspberry Pi + Amplifier + Coil





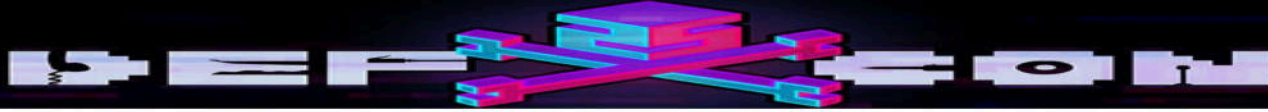


# Raspberry Pi - Demo



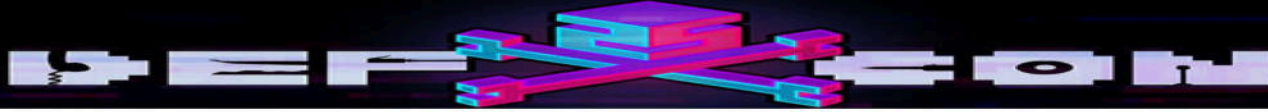
# Bluetooth Technology



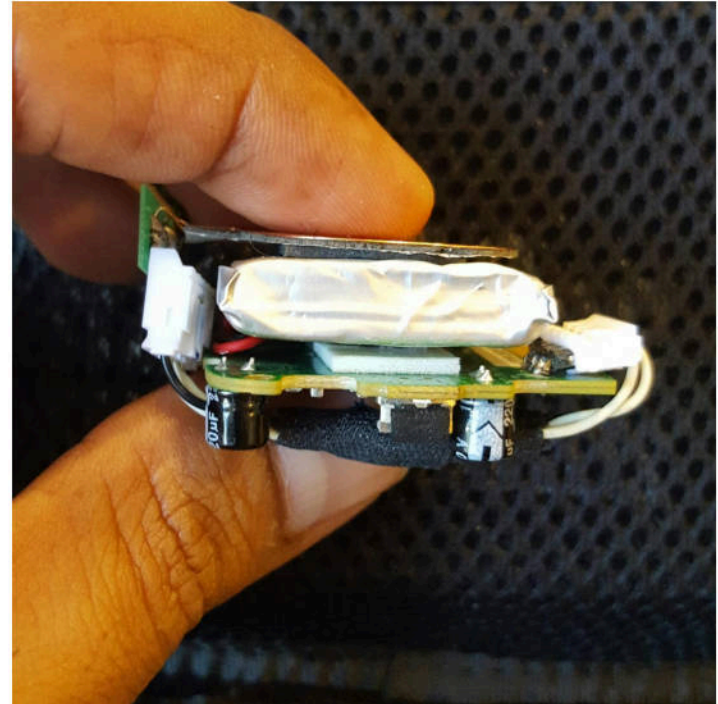


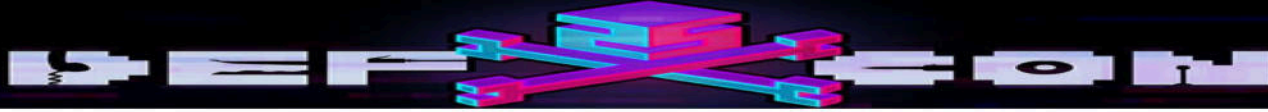
# Bluetooth Speaker





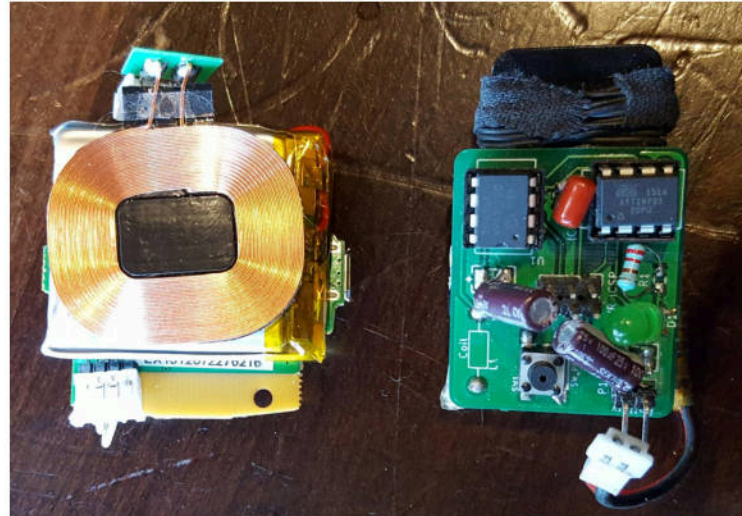
# Bluetooth Speaker





# MagSpoof Cousin

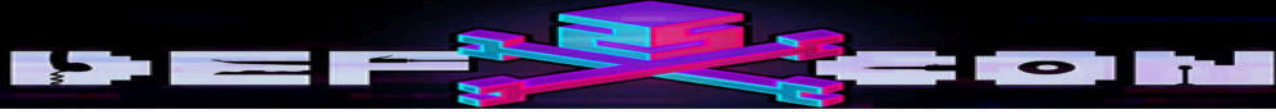
**BlueSpoof**



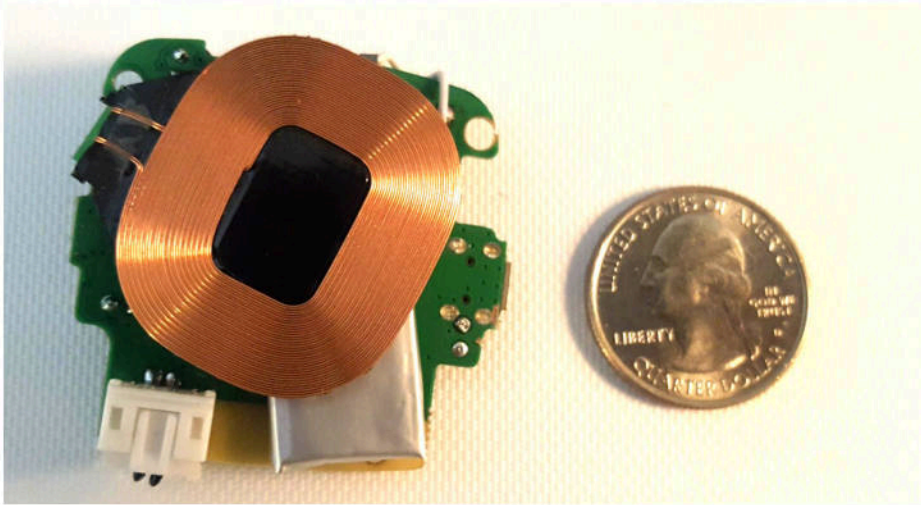
Electronic Cats(@electronicats)  
design

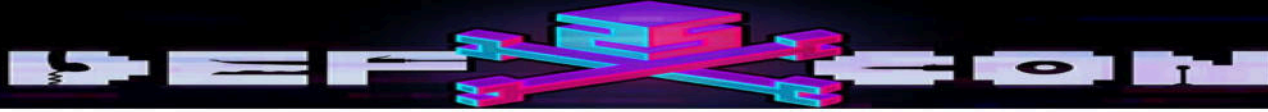
**MagSpoof**





# BlueSpooof

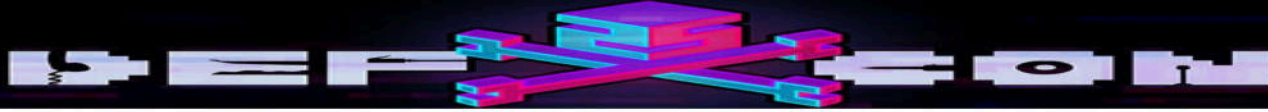




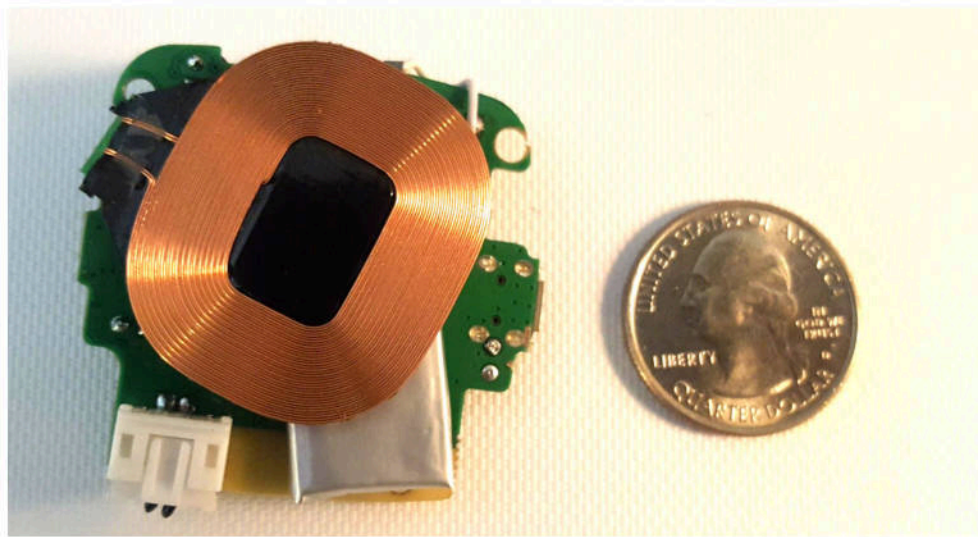
# BlueSpooof tool - Characteristics

- Cheap < \$20
- Easy to implement
- Escalable
- 3.7 V Battery
- Fast transmission
- Accurate

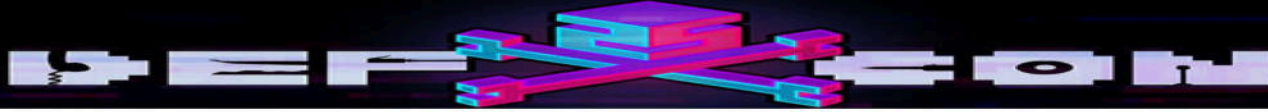




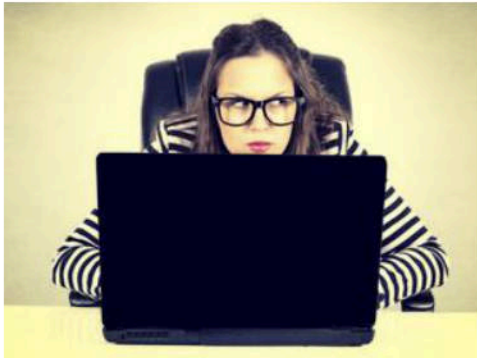
# BlueSpooof - Demo



<https://www.youtube.com/watch?v=elzqLhLnCek>



# Multiple Targets?

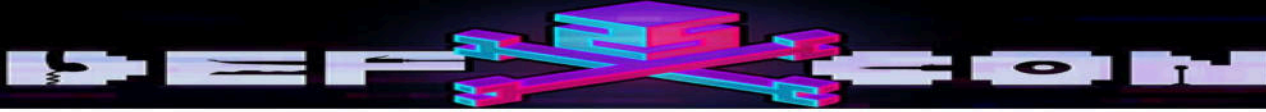


Token 1



Token 2





# Controlling Multiple Speakers?

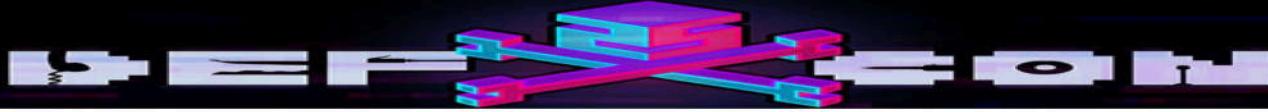
## Python Sound Device Library

```
[salvador@Sal-M:~/Desktop/defcon 25]$ python -m sounddevice
> 0 Built-in Microphone, Core Audio (2 in, 0 out)
  1 Built-in Output, Core Audio (0 in, 2 out)
  2 token-, Core Audio (1 in, 0 out)
  3 token-, Core Audio (0 in, 2 out)
  4 Marsboy, Core Audio (1 in, 0 out)
< 5 Marsboy, Core Audio (0 in, 2 out)
```

<https://pypi.python.org/pypi/sounddevice>

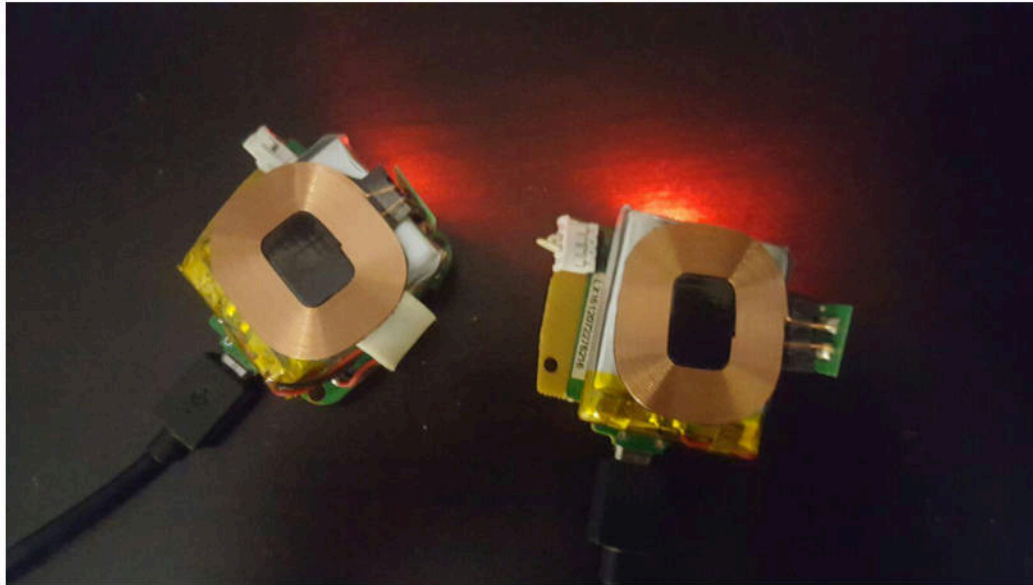






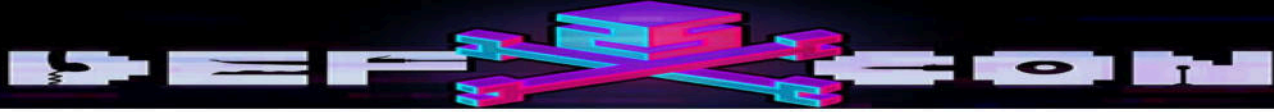
# Attack with Multiple Bluetooth Speakers?

---

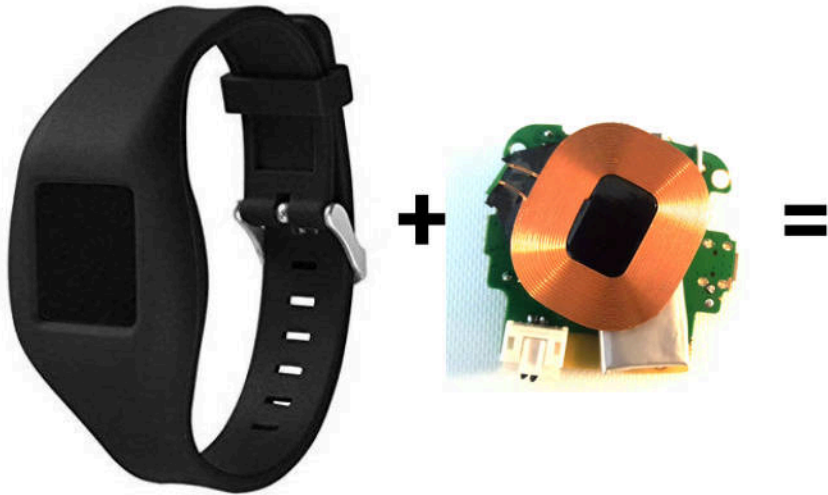


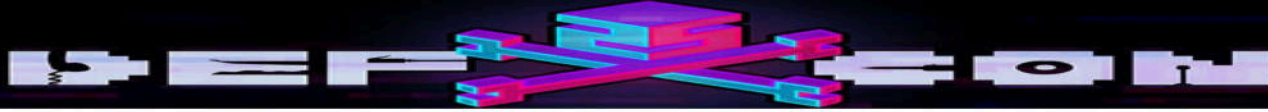
Demo

<https://www.youtube.com/watch?v=5hInVNLUC8s>

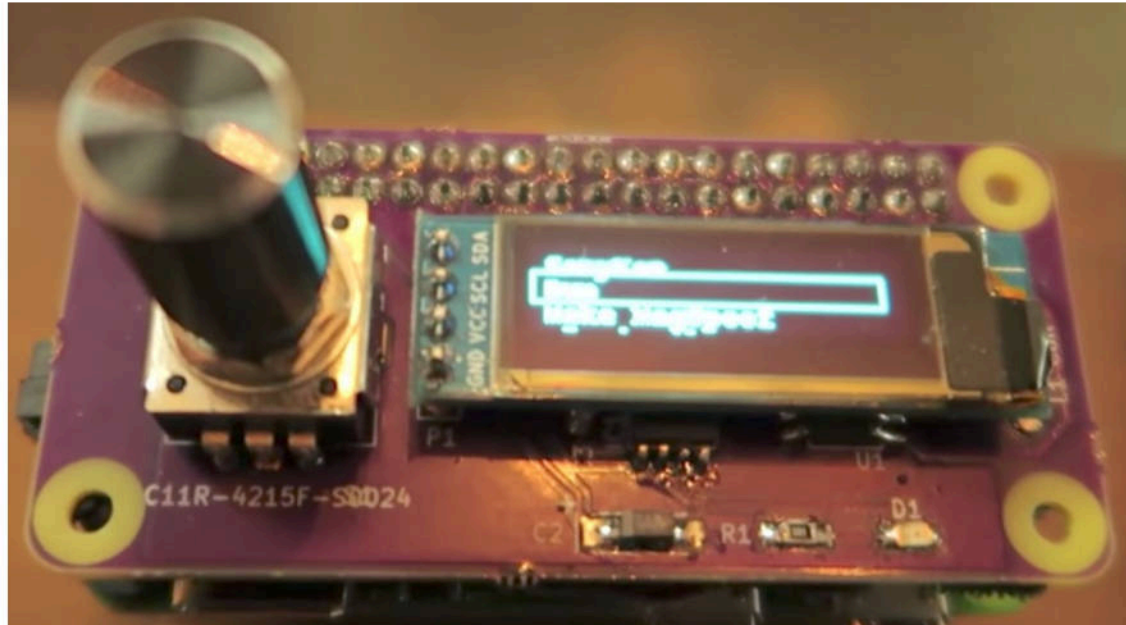


# Bonus Take-Away Project: iWey

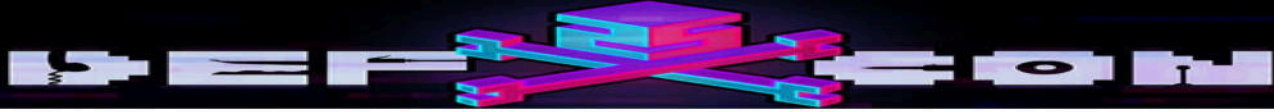




# SamyKam



<https://salmg.net/2017/01/16/samykam/>



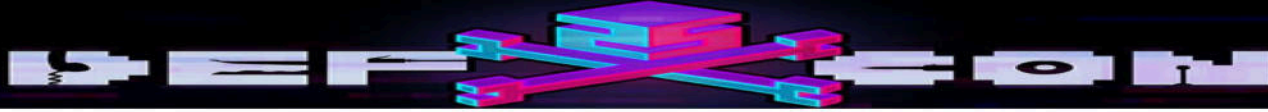
# Greetz, Hugs & Stuff

**Samy Kamkar (@samykamkar)**

**Electronic Cats (@electronicats)**

**RMHT (raza-mexicana.org)**

**Los Razos!**



# Questions?



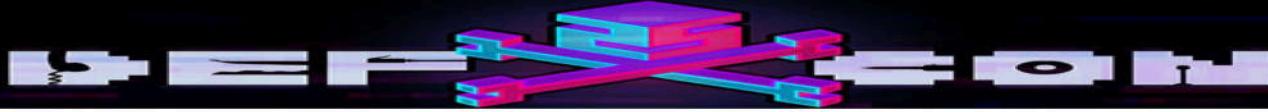
Salvador Mendoza

Twitter: [@Netxing](https://twitter.com/Netxing)

Blog: [salmg.net](http://salmg.net)

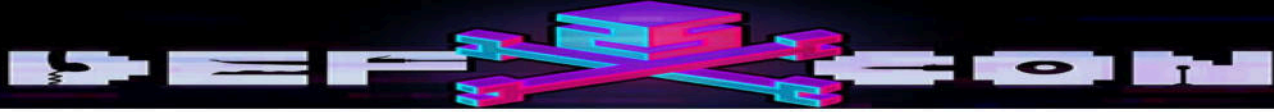
sal@salmg.net





**Thank you!**

**Happy Hacking  
Anniversary!**



# Resources

**Samy Kamkar: [samy.pl/magspooft](https://samy.pl/magspooft)**

**Electronic Cats: [twitter.com/electroniccats](https://twitter.com/electroniccats)**

**Major Malfunction: [youtube.com/watch?v=ITihB1c3dHw](https://youtube.com/watch?v=ITihB1c3dHw)**

**Weston Hecker: [youtube.com/watch?v=mV\\_0k9Fh590](https://youtube.com/watch?v=mV_0k9Fh590)**